VoiceOps 2026: ZentroMart Support Agent

Capstone implementation report | Individual submission | 22 October 2026

Public repository	https://github.com/priyam-chakraborty/voiceops-2026-zentromart
Core stack	Python 3.13.6, Gemini API for generation and embeddings, local document retrieval, supplied synthetic data/tools, Gnani.ai STT and TTS.

1. Problem and solution

ZentroMart is a fictional Indian online store whose support team receives recurring questions about delivery, returns, refunds, cancellations, and warranties. This project builds a support assistant intended to answer from company documents, handle selected multi-turn conversations, verify callers before revealing order details, escalate cases when required, and support voice input/output.

The application uses Gemini for language generation and embeddings, retrieves sections from the supplied knowledge base, uses the course-provided synthetic order data and mock tools, and uses Gnani.ai for both speech-to-text (STT) and text-to-speech (TTS). The RAG index successfully embedded 84 company-document sections and saved a local index.

2. Implementation and tested behavior

Stage	Implementation and observed test
Step 1: Prompt-only LLM	Implemented and launched successfully. It produces answers without retrieving the company documents, so policy-specific answers can be incomplete or unsupported.
Step 2: RAG	Indexed 84 sections. The grocery question retrieved current policy sections RET-5, RET-2, and DWM-1, and returned the rule that grocery/perishable items are not returnable for change of mind while eligible item issues must be reported within 48 hours with photos. An unsupported lifetime screen-replacement claim was not confirmed; current warranty terms were provided instead.
Step 3: Conversation and tools	Manual tests demonstrated caller verification with the last four phone digits, order lookup, and support-ticket creation. In the synthetic damaged-TV scenario, the assistant identified the INR 31,649 item as a high-value claim, refused to approve a replacement itself, retrieved DWM-3 and ESC-1/ESC-2, and reported a ticket only after the mock tool confirmed success. These examples do not establish that all 23 practice conversations pass.
Step 4: Voice	Installed the official gnani-vachana SDK and successfully tested Gnani STT and TTS. The batch runner processed two audio inputs successfully. A generated audio question about grocery returns completed transcription, document retrieval, answer generation, evidence capture, and WAV generation; the reply WAV was played and verified as audible.

3. Preliminary evaluation

The provided evaluation helper was run on five of the 52 practice questions. Its string/behavior heuristic reported 2/5 Step 1 passes (40%) and 3/5 Step 2 passes (60%). For Step 2, the helper reported gold knowledge-base evidence hits for 2/2 cases in which gold KB section IDs were specified. These are preliminary heuristic results only, not official course scores or a reliable estimate of unseen-test accuracy.

The evaluation was interrupted by API-provider errors. One Step 2 request returned HTTP 503 (temporary unavailability), and subsequent Gemini calls returned HTTP 429 (free-tier request quota exhausted). Results for affected calls are incomplete. The 23 supplied multi-turn practice conversations have not been systematically evaluated.

4. Problems, fixes, and limitations

The key operational limitation was Gemini rate/quota availability; repeated retries were stopped when the provider reported quota exhaustion. Retrieval for high-value damaged-item escalation was adjusted so required escalation-policy sections DWM-3 and ESC-1/ESC-2 appeared in evidence for the tested scenario. The development laptop had no microphone, so synthetic speech was generated with Gnani TTS and used as batch-runner input; this validates the technical pipeline but is not a test of live microphone capture or spontaneous human speech.

The system has been validated on selected manual examples, not comprehensively across the full practice set. A clean-machine reproduction check and final submission inspection remain important. The observed successful voice run confirms the tested path only, not production readiness or guaranteed performance on unseen cases.

5. Next improvements

Next priorities are to evaluate more practice questions when API quota allows, systematically test multi-turn follow-ups and tool outcomes, verify the exact batch command from a clean setup, and inspect the public repository and final ZIP for reproducibility and secret leakage. The system uses synthetic course data only; Gnani.ai is used for both STT and TTS.

6. Submission evidence

Item	Evidence observed
Dataset validation	Validator passed for all 9 CSVs, 21 knowledge-base documents, key relationships, practice evidence, verification gates, and company naming.
Grounded policy response	Saved sample output includes RET-5, RET-2, and DWM-1 for the grocery return question.
Voice output artifacts	Sample artifacts include transcript, reply text, result JSON, and a playable reply WAV.
Repository	Intended public repository: https://github.com/priyam-chakraborty/voiceops-2026-zentromart. Final contents and remote upload should be verified before submission.

Declaration of scope: Results above describe only the runs and manual tests observed during development. No claim is made that the full 52-question set or all 23 practice conversations passed.